

Merge Sort Dry Run Task (For Beginners)

Instructions

1. Take the array: **{38, 27, 43, 3, 9, 82, 10}**
 2. Follow the **Divide and Conquer** approach.
 3. First, divide the array until each group contains only one element.
 4. Then merge the groups back in sorted order.
 5. Write every step clearly.
 6. Do not jump directly to the final answer.
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Part A: Divide the Array

Complete the blanks.

Original Array:

{38, 27, 43, 3, 9, 82, 10}

Step 1:

{38, 27, 43} {3, 9, 82, 10}

Step 2:

{38} {27, 43} {3, 9} {82, 10}

Step 3:

{38} {27} {43} {3} {9} {82} {10}

Hint

Keep dividing each group into two smaller groups until only one element remains.

Part B: Merge the Left Half

Merge:

{27} and {43}

Result:

{27, 43}

Hint

Compare 27 and 43. Put the smaller number first.

Now merge:

{38} and {27, 43}

Fill the comparisons:

Compare 38 and 27 → Take 27

Compare 38 and 43 → Take 38

Take remaining 43

Result = {27,38,43}

Part C: Merge the Right Half

Merge:

{3} and {9}

Result:

{3,9}

Merge:

{82} and {10}

Result:

{10,82}

Hint

Always place the smaller element first.

Now merge:

{3, 9} and {10, 82}

Fill the steps:

Compare 3 and 10 → Take 3

Compare 9 and 10 → Take 9

Take remaining 10

Result = {3,9,10}

Part D: Final Merge

Merge:

{27, 38, 43}
and
{3, 9, 10, 82}

Fill in:

Compare 27 and 3 → Take 3

Compare 27 and 9 → Take 9

Compare 27 and 10 → Take 10

Compare 27 and 82 → Take 27

Compare 38 and 82 → Take 38

Compare 43 and 82 → Take 43

Take remaining 82

Final Sorted Array:

{3,9,10,27,38,43,82}

Teacher's Quick Hint

Remember:

Merge Sort = Divide → Divide → Divide → Merge → Merge → Merge

At every merge step:

“Compare the first elements of both groups and choose the smaller one.”

This is the only rule students need to remember during the dry run. 

